

Configuration Automation using Ansible

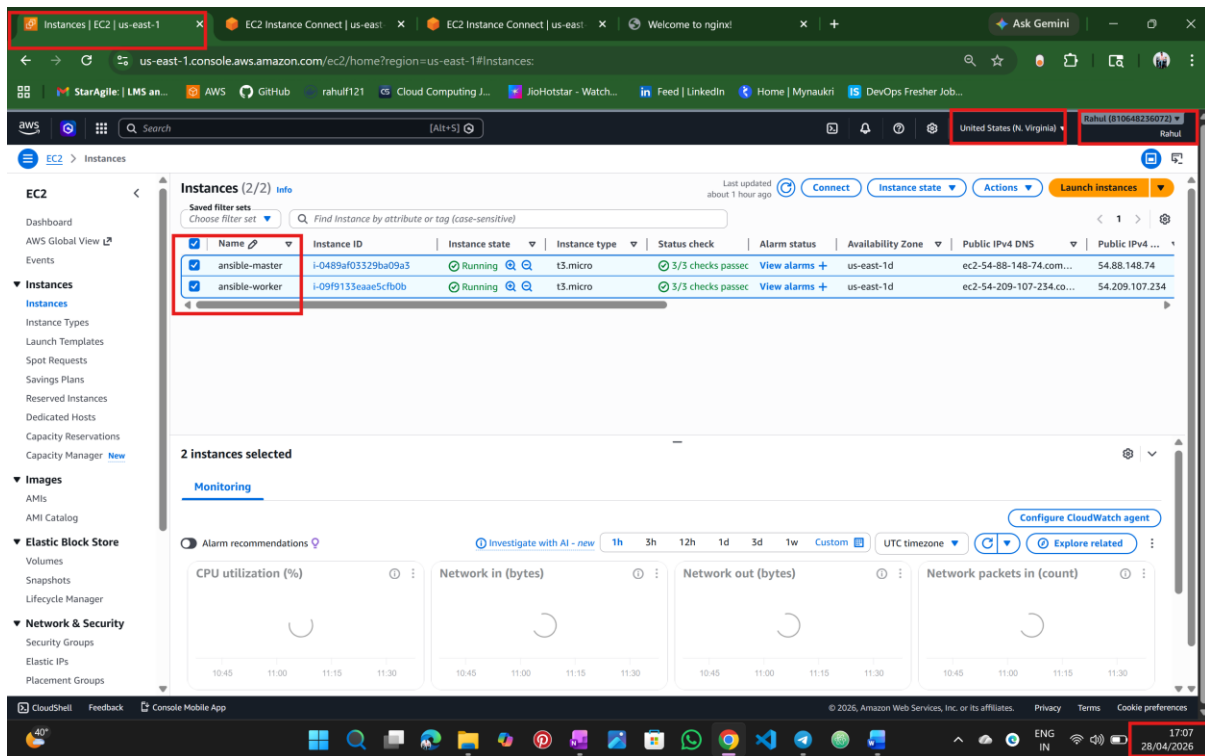
Submitted to - Vikul Sir

Date of Submission – 28/APR/2026

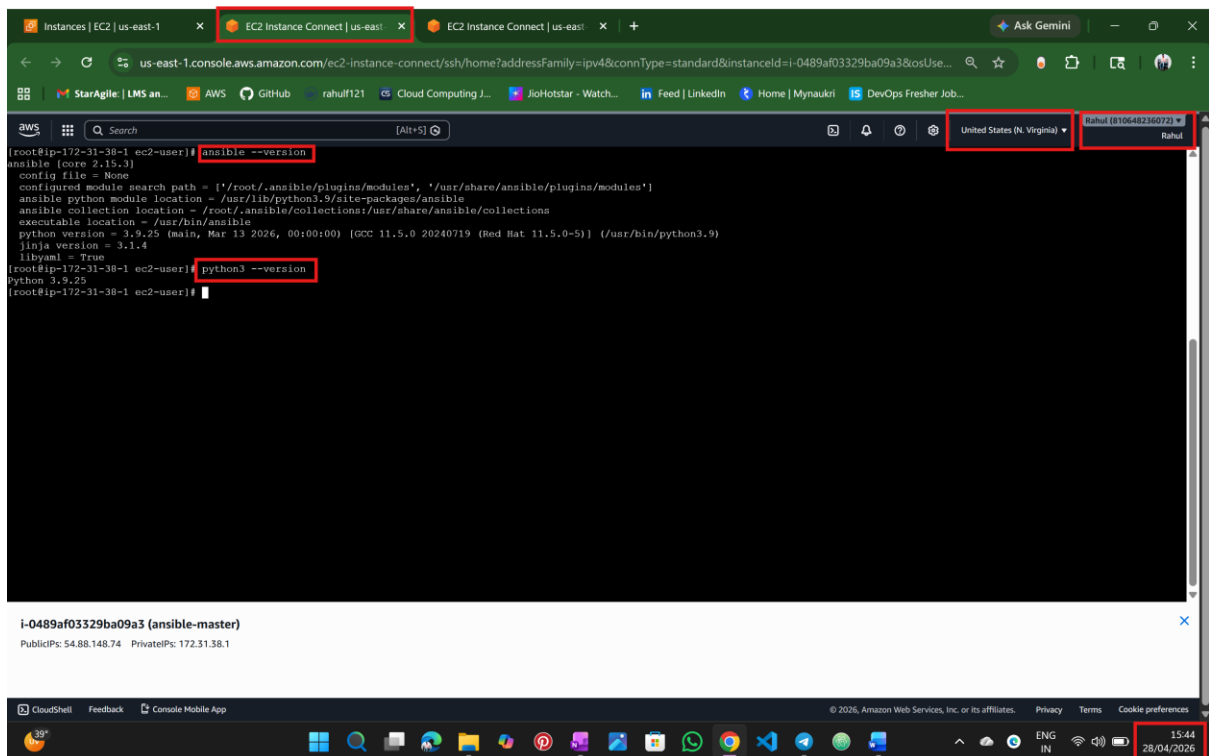
Submitted by – Rahul Nalathathi

L3 - Create Ansible Role to define the task, handler for Nginx Service Installation and invoke the role in Ansible playbook

Step :1 login into AWS console and create two ec2 instances



Step :2 This is ansible master node

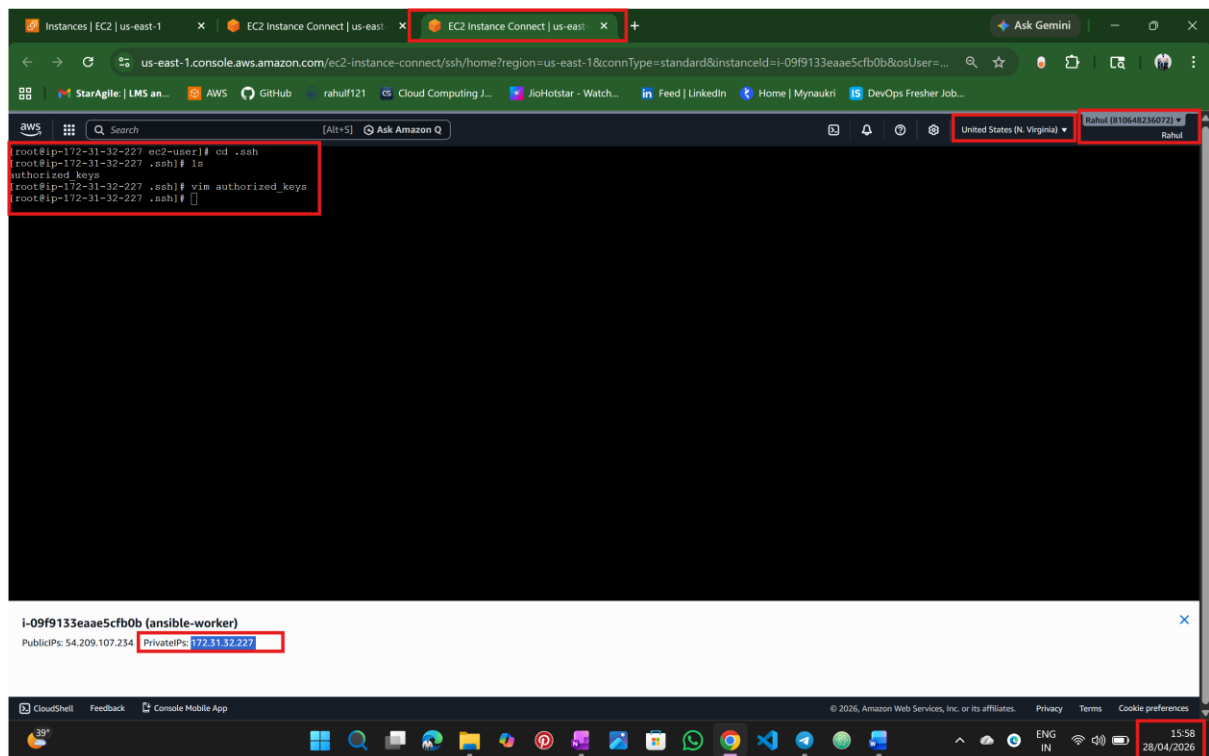


The screenshot shows the AWS CloudShell interface for the 'ansible-master' node. The terminal window displays the output of the 'ansible --version' and 'python3 --version' commands. The 'ansible --version' output shows the configuration file, search path, module location, collection location, executable location, python version (3.9.25), and Jinja version (3.1.4). The 'python3 --version' output shows 'Python 3.9.25'. The terminal window is titled 'i-0489af03329ba09a3 (ansible-master)' and shows the public IP as 54.88.148.74 and private IP as 172.31.38.1. The browser tabs at the top show 'EC2 Instance Connect | us-east-1' and 'EC2 Instance Connect | us-east-1'. The bottom status bar shows the time as 15:44 on 28/04/2026.

```
[root@ip-172-31-38-1 ec2-user]# ansible --version
ansible [core 2.15.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /root/.ansible/collectonst/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Mar 13 2026, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[root@ip-172-31-38-1 ec2-user]# python3 --version
Python 3.9.25
[root@ip-172-31-38-1 ec2-user]#
```

i-0489af03329ba09a3 (ansible-master)
PublicIPs: 54.88.148.74 PrivateIPs: 172.31.38.1

Step :3 This is ansible worker node

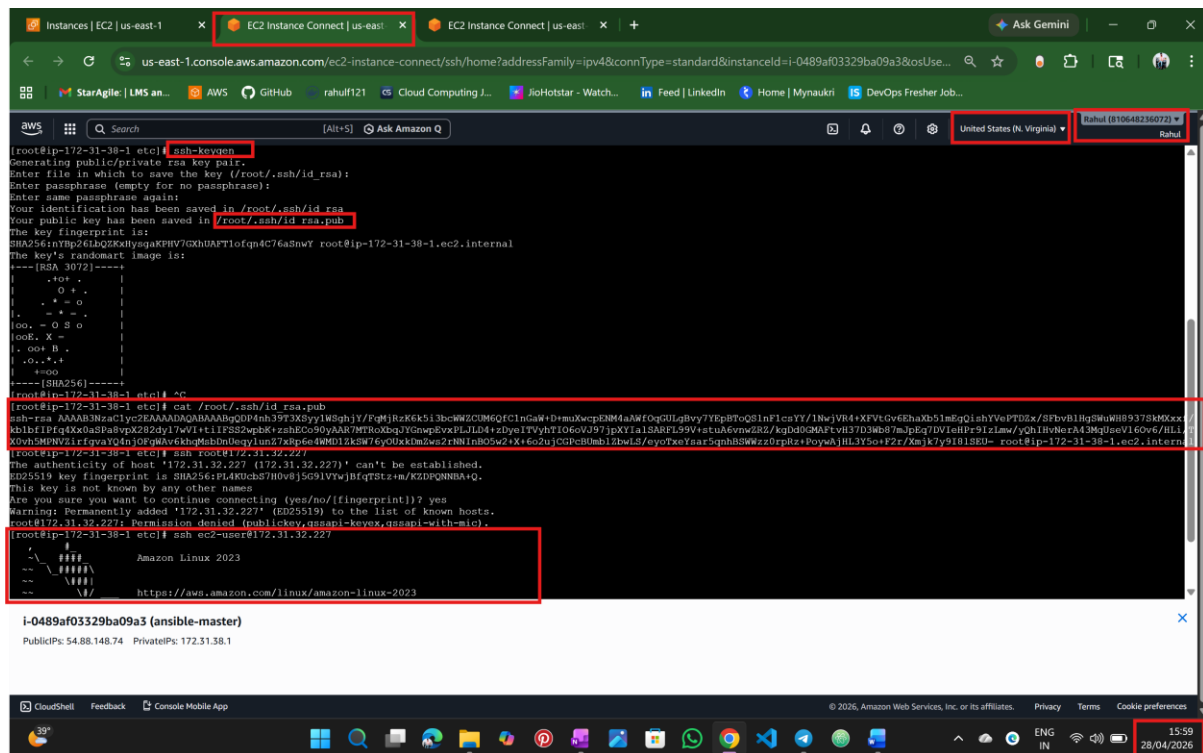


The screenshot shows the AWS CloudShell interface for the 'ansible-worker' node. The terminal window displays the output of the 'cd .ssh', 'ls', and 'vim authorized_keys' commands. The 'cd .ssh' command shows the current directory is '/root/.ssh'. The 'ls' command shows the contents of the directory, including 'authorized_keys'. The 'vim authorized_keys' command opens the file in the vim editor. The terminal window is titled 'i-09f9133eaa5cfb0b (ansible-worker)' and shows the public IP as 54.209.107.234 and private IP as 172.31.32.227. The browser tabs at the top show 'EC2 Instance Connect | us-east-1' and 'EC2 Instance Connect | us-east-1'. The bottom status bar shows the time as 15:58 on 28/04/2026.

```
[root@ip-172-31-32-227 ec2-user]# cd .ssh
[root@ip-172-31-32-227 .ssh]# ls
authorized_keys
[root@ip-172-31-32-227 .ssh]# vim authorized_keys
[root@ip-172-31-32-227 .ssh]#
```

i-09f9133eaa5cfb0b (ansible-worker)
PublicIPs: 54.209.107.234 PrivateIPs: 172.31.32.227

Step :4 Connect ansible master node and ansible worker node with ssh-keygen



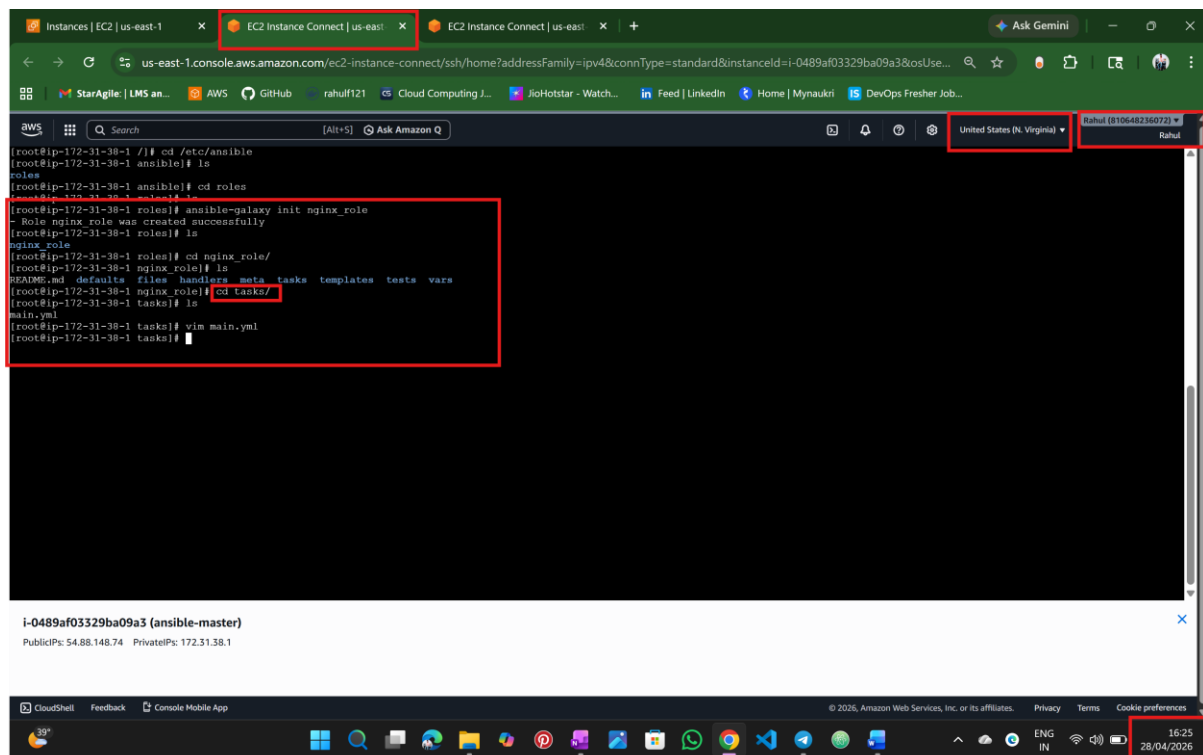
The screenshot shows the AWS CloudShell interface. The terminal window displays the following commands and output:

```
[root@ip-172-31-38-1 etc]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:nYBp26LbQZKkHlygqKPHV7GKhUAP7lofq4C76aSnwY root@ip-172-31-38-1.ec2.internal
The key's randomart image is:
+-----[RSA 3072]-----+
|
| ..+..
|  O + .
|  + + +
| = + +
|
|oo. = O S o
|oeX. X
|  ooa B
|  .o..+.
|   +
|  +
+-----[SHA256]-----+
[root@ip-172-31-38-1 etc]# cat /root/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQDAh39T3X5y1W5ghjY/FqHjRzK6k5i3b-cWZCUM6GfC1nGaW+DmuXwcpENM4aWfoQgULg9v7YEp8ToQsInF1cnyY/1Nw1VB4+XfVtGvEhaxb51mgQishYVvePTD5x/SFbvB1Hg5WuH89373kMXck/
X0vh5MhV2ifrguYQ4nJOEgKAv6kHmMdbnUeqylun27Kp6e4NM012kSM76Y0UxkDm2w2rNNinB05w24x6o2uJCGPCBmb12wLS/eyoTxeYsar5qhb5Wwz0rpRz+PoywhjHL3Y5o+F2r/Xm1k7y91B18ED= root@ip-172-31-38-1.ec2.internal
[root@ip-172-31-38-1 etc]# ssh root@172.31.32.227
Warning: Permanently added '172.31.32.227' (ED25519) to the list of known hosts.
Permission denied (publickey,assh1-key,assh1-with-mic).
[root@ip-172-31-38-1 etc]# ssh ec2-user@172.31.32.227
Warning: Permanently added '172.31.32.227' (ED25519) to the list of known hosts.
ec2-user@ip-172-31-32-227:~$
~$ cat /etc/ansible/hosts
172.31.32.227 ansible_host=172.31.32.227
~$ cd /etc/ansible
~$ ls
roles
~$ cd roles
~$ ansible-galaxy init nginx_role
- Role nginx_role was created successfully
~$ cd nginx_role/
~$ ls
README.md defaults files handlers meta tasks templates tests vars
~$ cd tasks/
~$ ls
main.yml
~$ vim main.yml
```

The terminal output shows the successful generation of an SSH key pair and the connection to the EC2 instance. The connection is successful, and the user is prompted to enter a password. The user enters the password, and the connection is established. The user is then prompted to enter a command, and the user enters 'cat /etc/ansible/hosts'. The output shows the contents of the hosts file, which contains the IP address of the EC2 instance and the ansible_host variable.

Step :5 Now, create roles

→for tasks



The screenshot shows the AWS CloudShell interface. The terminal window displays the following commands and output:

```
[root@ip-172-31-38-1 /]# cd /etc/ansible
[root@ip-172-31-38-1 ansible]# ls
roles
[root@ip-172-31-38-1 ansible]# cd roles
[root@ip-172-31-38-1 roles]# ansible-galaxy init nginx_role
- Role nginx_role was created successfully
[root@ip-172-31-38-1 roles]# cd nginx_role/
[root@ip-172-31-38-1 nginx_role]# ls
README.md defaults files handlers meta tasks templates tests vars
[root@ip-172-31-38-1 nginx_role]# cd tasks/
[root@ip-172-31-38-1 tasks]# ls
main.yml
[root@ip-172-31-38-1 tasks]# vim main.yml
```

The terminal output shows the successful creation of the 'nginx_role' role. The user is then prompted to enter a command, and the user enters 'cd tasks/'. The user is then prompted to enter a command, and the user enters 'ls'. The output shows the contents of the tasks directory, which contains the 'main.yml' file. The user is then prompted to enter a command, and the user enters 'vim main.yml'.

The screenshot shows the AWS CloudShell interface with a terminal window. The terminal displays the following Ansible tasks for the `nginx_role`:

```
--# tasks file for nginx_role
--
-- name: Install Nginx
--   yum:
--     name: nginx
--     state: present
--     notify: Restart Nginx
-- name: Start and Enable Nginx
--   service:
--     name: nginx
--     state: started
--     enabled: yes
```

The terminal also shows the instance ID `i-0489af03329ba09a3` (ansible-master) and its public/private IP addresses. The AWS console header shows the region as `United States (N. Virginia)` and the user as `Rahul`. The bottom status bar shows the time as `16:25` on `28/04/2026`.

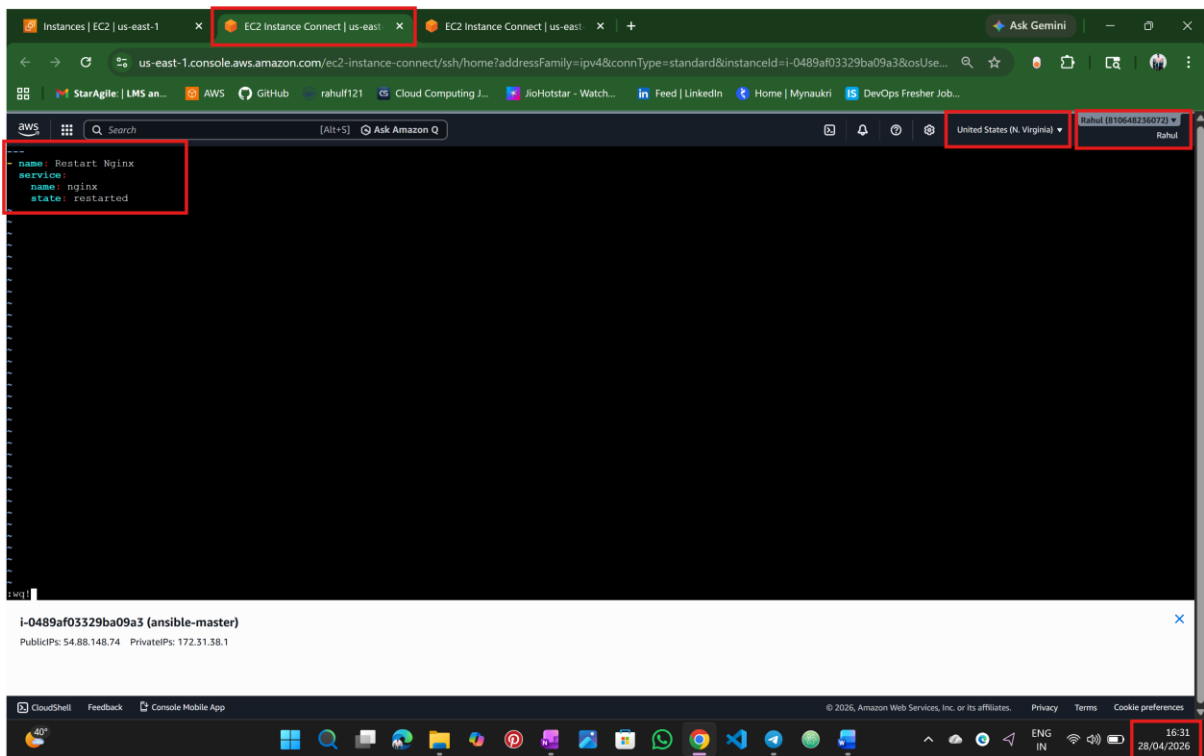
Step :6 create roles

→for handlers

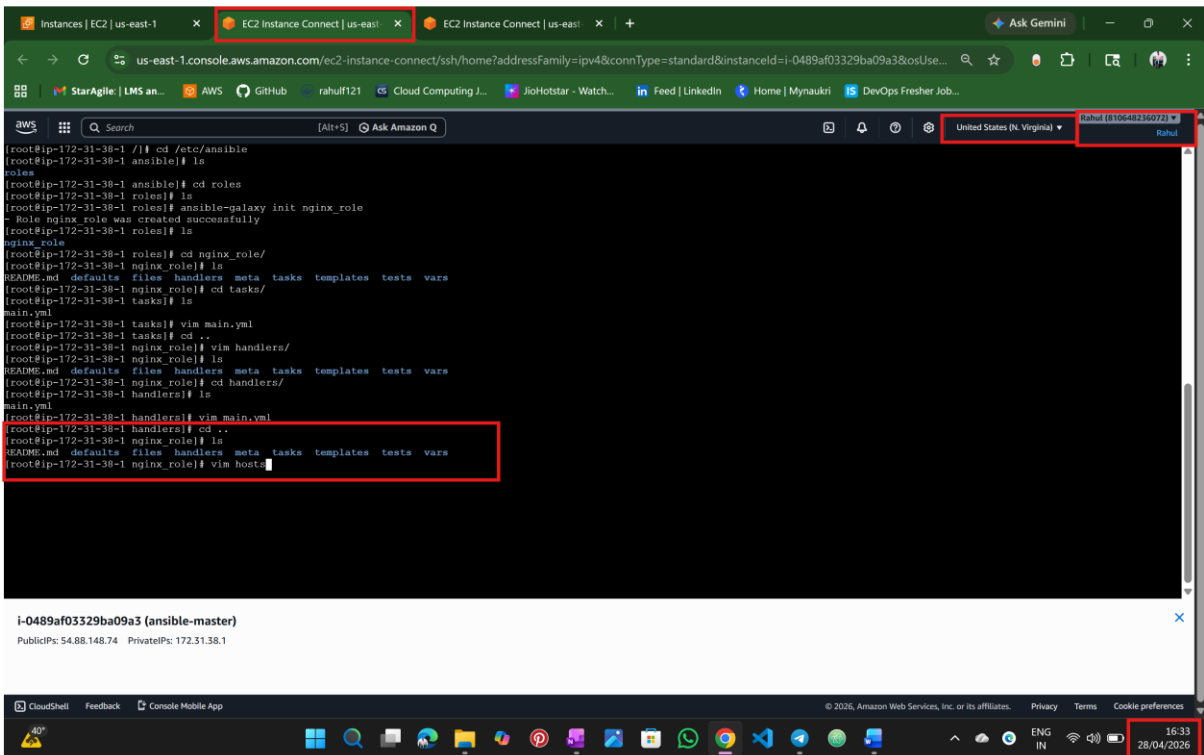
The screenshot shows the AWS CloudShell interface with a terminal window. The terminal displays the following commands and output for creating the `nginx_role` and its handlers:

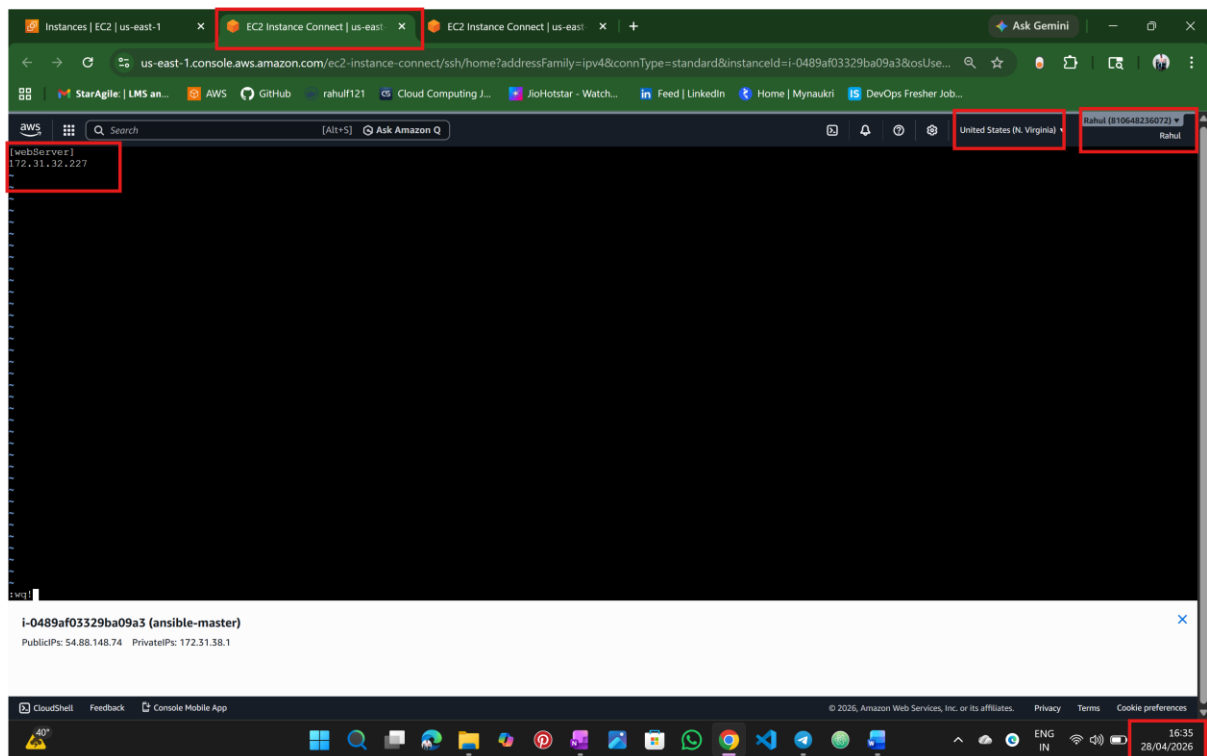
```
(root@ip-172-31-38-1 /) # cd /etc/ansible
[root@ip-172-31-38-1 ansible] # ls
roles
[root@ip-172-31-38-1 ansible] # cd roles
[root@ip-172-31-38-1 roles] # ls
ansible-galaxy init nginx_role
- Role nginx_role was created successfully
[root@ip-172-31-38-1 roles] # ls
nginx_role
[root@ip-172-31-38-1 roles] # cd nginx_role/
[root@ip-172-31-38-1 nginx_role] # ls
README.md defaults files handlers meta tasks templates tests vars
[root@ip-172-31-38-1 nginx_role] # cd tasks/
[root@ip-172-31-38-1 tasks] # ls
main.yml
[root@ip-172-31-38-1 tasks] # vim main.yml
[root@ip-172-31-38-1 tasks] # cd ../
[root@ip-172-31-38-1 nginx_role] # vim handlers/
[root@ip-172-31-38-1 nginx_role] # ls
README.md defaults files handlers meta tasks templates tests vars
[root@ip-172-31-38-1 nginx_role] # cd handlers/
[root@ip-172-31-38-1 handlers] # ls
main.yml
[root@ip-172-31-38-1 handlers] # vim main.yml
```

The terminal also shows the instance ID `i-0489af03329ba09a3` (ansible-master) and its public/private IP addresses. The AWS console header shows the region as `United States (N. Virginia)` and the user as `Rahul`. The bottom status bar shows the time as `16:29` on `28/04/2026`.

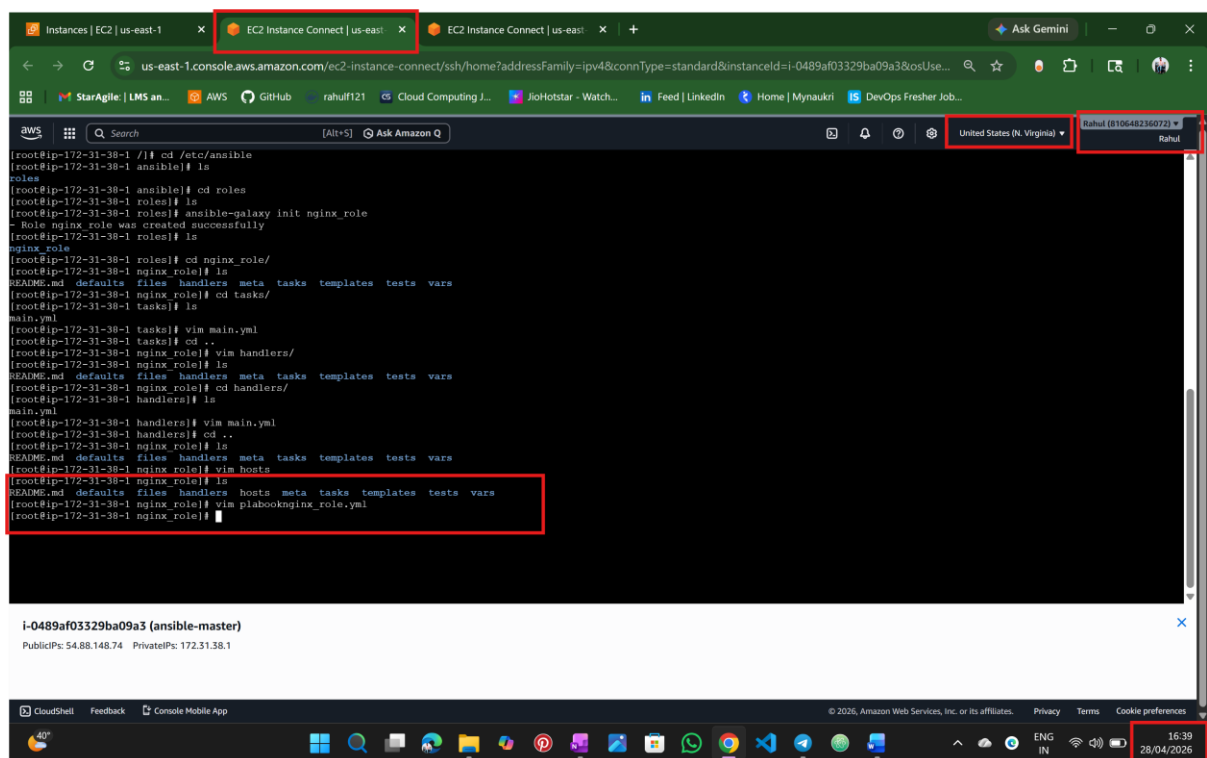


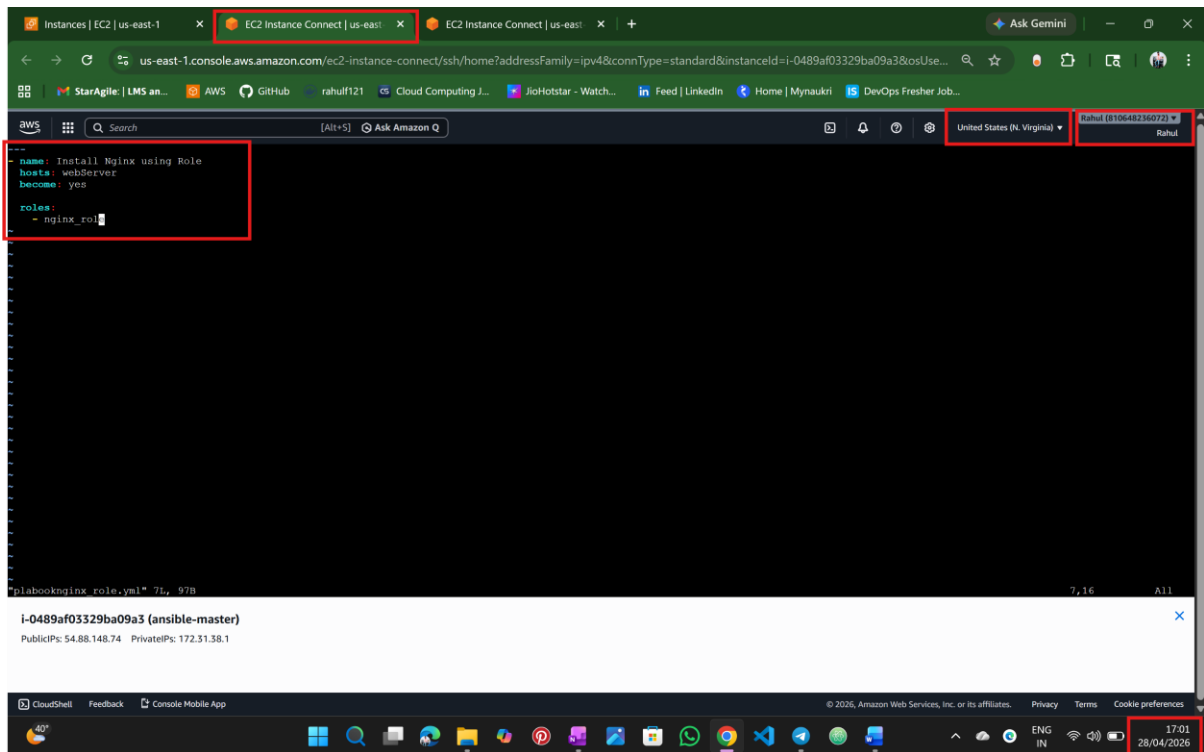
Step :7 create inventory file



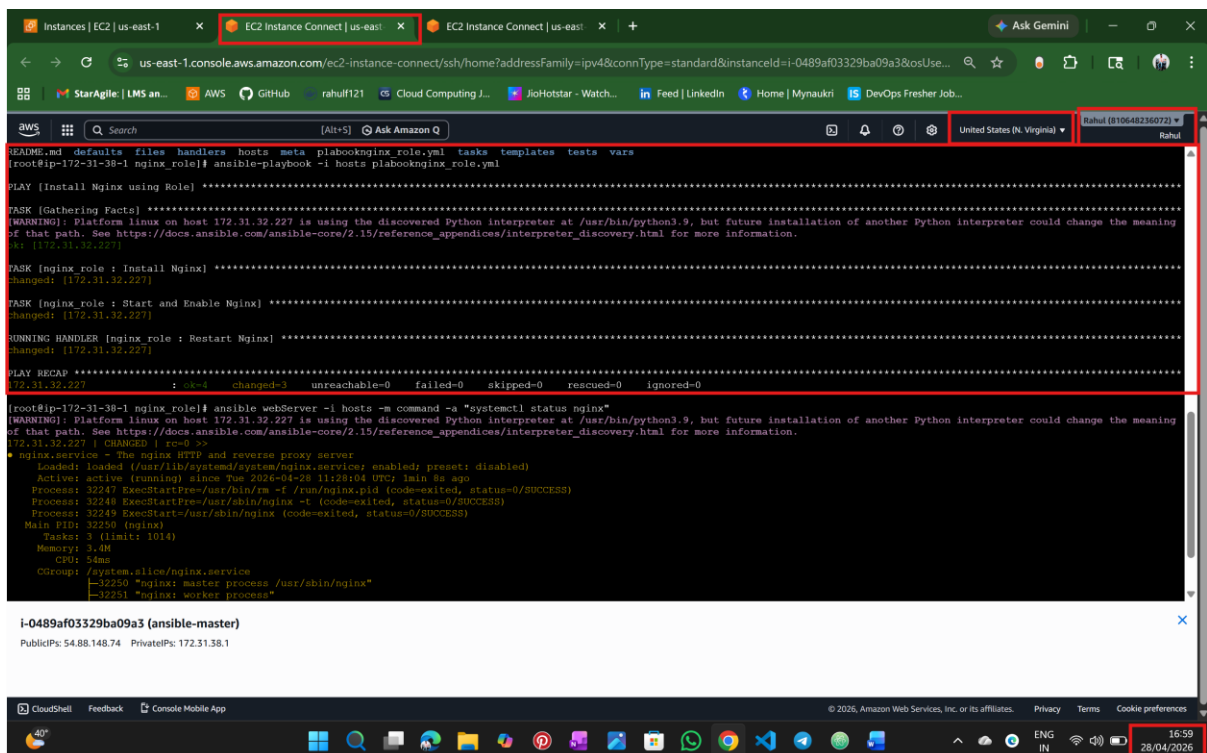


Step :8 create playbook

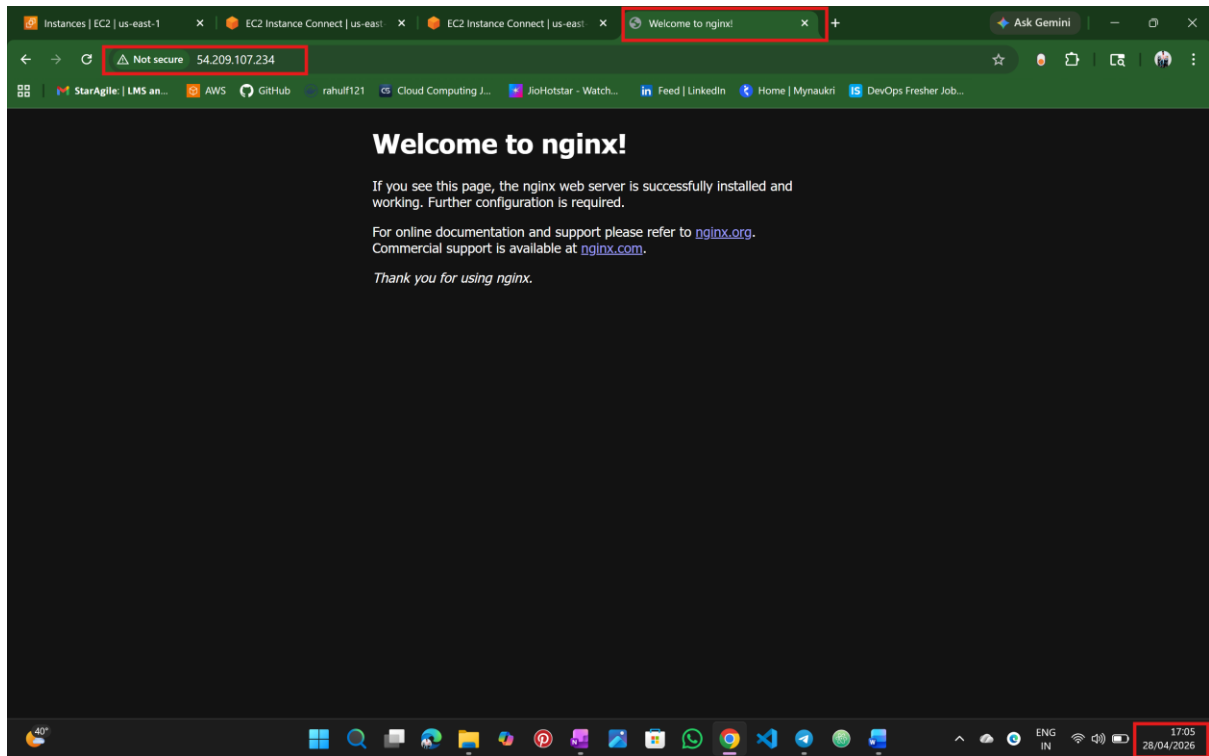




Step :9 Run the Playbook



Step :10 Access the application via pubic ip of the worker node



Nalathati Rahul